

"I feel you, but...": Exploring the Boundaries of Pragmatic Markers in Real-time Voice-based Conversational Agents

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Abstract

Although Voice-based Conversational Agents (VCAs) continue to evolve, achieving natural interaction remains challenging. While studies have explored various aspects of conversational style, the impact of Pragmatic Markers (PMs)—linguistic elements that convey speaker intention and emotional context—remains underexplored in human-agent interactions. We conducted a formative study with eight participants using a prototype VCA system, which revealed that real-time sentiment analysis could support more effective PM usage by maintaining conversational context. Building on these insights, we developed a VCA system integrating sentiment-based PM selection and conducted a main study with 20 participants using a 2×2 within-subjects design (emotional/work-related topics × with/without markers). Analysis of conversation logs, surveys, and interviews revealed that moderate PM usage enhanced perceived naturalness and emotional engagement. However, excessive use negatively impacted user experience as participants expected VCAs to maintain a **passive yet attentive listener** role. Our findings inform design guidelines for context-aware PM integration in VCAs.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Conversational Agents, Pragmatic Markers, Real-time Interaction, Human-AI Interaction, Korean Language

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1 Introduction

Communication capabilities of Conversational Agents (CAs) have significantly improved [11, 19]. Accordingly, Voice-based Conversational Agents (VCAs) like Alexa [38] and Siri [32] have become familiar personal assistants in our daily lives, sparking research interest in understanding how humans perceive and interact with these systems [5, 34]. VCA applications have expanded into specific domains including education [9] and healthcare [44]. However, VCA systems still face notable challenges as the lack of emotional intelligence often leads to decreased user trust and degraded user experience [26, 27]. Technical improvements remain crucial, yet understanding how people engage with VCAs has become equally important for enhancing human-agent interactions [35, 41]. Current research shows that studies of speech-based interaction lag behind text-based counterparts [11, 33]. Given these challenges, this study focuses on how human-like linguistic features affect user experience and perception with VCAs in various contexts.

The Computers Are Social Actors (CASA) paradigm [29] suggests that human-computer interaction follows similar social rules as human-human interaction, making anthropomorphism a key design strategy for CAs [36]. In particular, human-like interaction characteristics and linguistic cues in CA interactions [40] positively influence user trust and acceptance [12, 23, 31, 42] and the absence of such characteristics negatively impacts user experience [42]. Notably, among various linguistic elements, conversational fillers have been studied in recent years [1, 2]. Jeong et al. [16] and Liu et al. [22] show that fillers affect perceived intelligence, humanness, and likability in conversation, while enhancing persuasiveness in marketing scenarios. However, current research on fillers has limitations, as they are typically considered simple lexical elements

(such as ‘um’, ‘uh’, and ‘well’), neglecting the contextual dynamics present in real-life conversations. We aim to explore more sophisticated linguistic cues that can enhance contextual communication in VCAs.

Among various linguistic cues in communication, Pragmatic Markers (PMs) constitute an important area of study [6, 43]. These markers are linguistic elements that, along with propositional content, help construct sentence meaning by signaling speaker intent and context [6]. Whereas fillers primarily signal hesitation or interruptions in speech [16], PMs function structurally within utterances to convey emotional nuances, interactional intentions, and contextual cues essential for interpreting conversational implicature [43]. Recent studies have demonstrated that appropriate use of PMs in conversational systems can enhance trust, relatability, and adaptiveness [25], while reducing cognitive load for users [39]. At the same time, inappropriate deployment of these markers can negatively impact user experience [18].

Despite their significance, current research on PMs remains largely limited to WoZ experiments and text-based scenarios [18, 25], leaving a gap in our understanding of how these markers function in real-time agent interactions. While sentiment-based dialogue control has been studied in CAs [10, 24], research on the emotional aspects of pragmatic markers is still limited, in spite of their critical role in conveying affective nuances in human conversation. Furthermore, research has been largely centered on English, leaving other languages relatively underexplored. This limitation is particularly problematic for high-context languages [8] like Korean, where contextual cues and interpersonal dynamics play a pivotal role [15], indicating that PMs might serve even more nuanced functions under these conditions.

In this study, we seek to explore how real-time interactions using conversationally relevant PMs influence communication in diverse contexts. We conducted a two-phase study investigating Korean PMs in VCA interactions. To explore the potential and limitations of Korean PMs, we first conducted a formative study with eight native Korean speakers. We developed a system combining a GPT-4o-based VA with Speech-to-Text (STT), Text-to-Speech (TTS) capabilities, utilizing a limited set of predefined markers. Participants engaged in at least five conversational turns with both marker-enabled and non-marker versions of the agent, followed by in-depth interviews to gather user experience feedback. While participants found the markers engaging and human-like, we identified three key challenges: misalignment between emotional context and markers, repetitive marker patterns, and turn-based system limitations.

Building on our formative study findings, we developed an enhanced system integrating the Hyper Clova X model [45] for real-time emotion recognition and OpenAI’s Realtime API [30] to implement natural and continuous voice conversations. The system automatically adjusts marker selection based on emotional context and ensures natural conversation flow. We conducted a 2×2 within-subjects study with 20 regular VCA users, where participants engaged in both emotional and work-related conversations with and without markers. Through analysis of conversation logs, surveys, and interviews, we found that markers had contrasting effects depending on the conversation context. They enhanced engagement and perceived empathy in emotional conversations, while

proving less effective or potentially disruptive in work-related interactions. This suggests that the effectiveness of pragmatic markers is highly context-dependent, highlighting the importance of selective marker implementation based on conversation type.

We contribute to the design of human-agent interactions in three key ways. First, we provide empirical evidence on how real-time PM implementation influences user perception and engagement in voice-based conversational agents. Second, we introduce the concept of the **passive yet attentive listener** as an ideal agent stance, maintaining a balance between conversational naturalness and appropriate agent role perception. Finally, we present design guidelines for context-aware PM integration in conversational agents, addressing both temporal dynamics and interaction context—factors that previous WoZ studies could not fully capture.

2 Formative Study

First, we investigate whether an LLM can naturally integrate PMs in ways that align with conversational context. We also sought to understand how participants would perceive these markers once they noticed them, aiming to gather insights into both their potential advantages and drawbacks. Finally, We conducted this study to inform the creation of a high-feasibility prototype tool, which will serve as the foundation for broader guideline exploration in a subsequent main study.

We constructed a GPT-4o-based VCA that processes user input through STT and generates an output whenever participants provide an input. This output was subsequently converted to speech through a TTS. To explore initial feasibility with a simplified set, we restricted the number of markers to a few predefined types and allowed GPT-4o to determine when and how to insert each marker. Conversations proceeded in a turn-based fashion, where participants would listen to one response and press a button to move on to the next turn.

The evaluation process included 8 participants (aged 19–64, native Korean speakers) and asked each participant to interact with both a marker-enabled agent and a non-marker agent, exchanging at least five conversational turns per condition. This was followed by a post-experiment interview to gather user experience feedback. The entire experiment lasted around 30 minutes. The semi-structured interviews were analyzed using thematic analysis with open coding, enabling us to identify core themes and insights regarding marker usage and overall user experience.

From the results, participants consistently recognized the markers and often remarked that they found the VCA more intriguing and human-like. Nonetheless, there were reports that the markers sometimes appeared misaligned with emotional context, creating impressions of forced or mechanical usage. Participants also highlighted discomfort with repeatedly hearing the same markers, which diminished the overall sense of naturalness. Additionally, the tonal quality of the TTS output occasionally overemphasized the markers, indicating a need for careful control of speed and pitch. While the turn-based structure made the markers more noticeable, it introduced a somewhat contrived feel to the interaction, suggesting that real-time conversation may be necessary for a more seamless, authentic user experience.

These findings provided key considerations for developing the VCA prototype for our main study. For the LLM component, our results suggested that incorporating sentiment analysis before marker selection could be beneficial, given the challenges observed in autonomous marker usage. The findings also pointed to the potential value of expanding and diversifying the marker inventory. Regarding the TTS component, our observations indicated that aligning tonal qualities with LLM-selected markers might enhance the naturalness of delivery. Additionally, the limitations of our turn-based system suggest that exploring real-time conversation capabilities could lead to more fluid interactions. These insights offer preliminary guidance for our main study’s development of an enhanced marker-enabled VCA.

3 Main Study

Building on our findings from the formative study, we designed our main study to enable real-time conversations between users and the system, incorporating sentiment analysis for marker selection during interactions. We expanded the marker inventory and refined the prompts based on prior Korean PM research to support more natural and contextually appropriate marker usage. To understand the practical impact of these improvements, we incorporated different conversation conditions to examine how they influence user interactions with the system.

3.1 System Design and Implementation

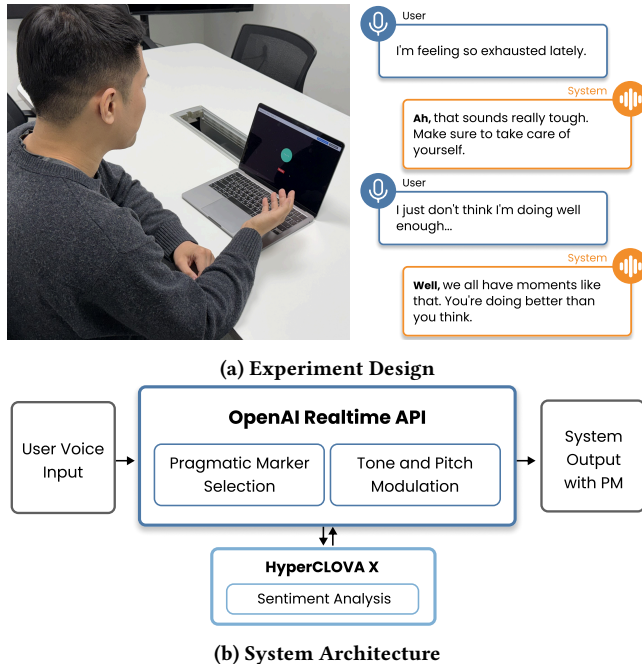


Figure 1: (a) Experiment Design: User interacting with the system in a work environment. (b) System Architecture: Components include sentiment analysis (HyperCLOVA X), tone modulation and pragmatic marker selection via OpenAI’s API to generate empathetic, context-aware responses.

Based on the design requirements identified in our formative study, we developed an improved system architecture that addresses the key challenges in real-time emotion recognition and the implementation of natural PMs (Figure 2). The system integrates two primary components: A sentiment analysis tool based on the Hyper Clova X model [45] and OpenAI’s Realtime API [30], which enables bidirectional speech-to-speech conversations with low latency and maintains natural interaction patterns. To accurately capture Korean cultural and linguistic nuances, we implemented sentiment analysis tool powered by the HyperClova X model as an API endpoint, specifically chosen for its sophisticated understanding of Korean language patterns and emotional expressions. The Realtime API-based agent from OpenAI was selected to maintain conversation fluency while ensuring high-quality responses with prosody-aware speech output. The system prompt was developed referencing prior works [13, 14, 20]. Its detailed configuration is provided in Appendix A.

The integration of these components creates a dynamic system where the real-time agent utilizes tool calling to invoke the sentiment analysis tool, which classifies user emotions into positive, neutral, or negative categories. Based on this emotional assessment, the system automatically adjusts both its conversational tone and the selection of PMs from an expanded marker inventory, which includes various discourse functions such as topic transitions, emphasis, agreement, and hesitation. The Realtime API agent dynamically adjusts its tonal qualities based on the selected markers and emotional context via predefined prompts, incorporating fine-tuned parameters for pitch, speed, and emphasis. This context-aware approach enables more natural interactions by preventing marker repetition and ensuring acoustic properties align with intended pragmatic functions.

3.2 Study Design

This study’s participants consisted of 20 native Korean speakers (aged 19–64), with a balanced gender distribution and diverse educational backgrounds (see Appendix B for detailed demographics). Following methodological precedents in VCA research [4, 17, 21], this sample size has been shown to be adequate for detecting significant interaction effects in VCA evaluation studies. Participants were recruited through online advertisements and institutional mailing lists, with the requirement of regular conversational agent usage (at least once a week) within the past six months. All participants underwent screening for hearing and speech capabilities to ensure proper interaction with the Korean voice interface. Participants’ technical proficiency was assessed using a 7-point Likert scale, with 75% showing moderate to high familiarity (4 or higher) with conversational agents. As part of our blinding procedure to minimize response bias, participants were not informed about the study’s specific focus on PMs. Instead, during recruitment, instruction, and throughout the experimental procedure, participants were only told that the study investigated general experiences with VCA conversations.

The study employed a 2 (topic) × 2 (pragmatic marker) within-subjects design, where each participant completed four distinct conversation sessions. Each session was designed to last between

three and five minutes ($M = 4.24$ minutes, $SD = 1.04$ minutes), providing sufficient interaction time while preventing fatigue. The conversations covered two primary domains: emotional topics addressing personal experiences and relationships, and work-related topics focusing on professional experiences and workplace scenarios [3]. These topics were carefully selected to elicit natural dialogue across different contexts while maintaining consistent complexity levels. Within each topic domain, conversations were conducted both with and without pragmatic markers, allowing for systematic comparison of interaction patterns. The task design underwent pilot testing with three participants who did not take part in the main study to validate timing and comprehensibility.

The IRB-approved procedure employed a structured 70-minute protocol divided into three main phases. The first phase consisted of four conversation-survey cycles (40 minutes total), where each participant engaged in the four experimental conditions. To minimize priming effects, participants were randomly assigned to different condition sequences following a balanced Latin square design, ensuring that each condition appeared equally across ordinal positions. Each conversation session (three-five minutes) was immediately followed by a five-minute post-session survey to capture fresh impressions while minimizing memory effects. Between sessions, participants were given short breaks to prevent fatigue and maintain engagement quality. The second phase comprised a comprehensive 30-minute semi-structured interview designed to gather detailed qualitative feedback about participants' experiences across all conditions. As a strategy to address potential biases, we adopted a structured reflection approach, initially asking participants to describe any differences they noticed between sessions. After they had fully shared their experiences, we progressively disclosed the primary research objective regarding PM and gathered their thoughts on the topic. All sessions were recorded and transcribed for data analysis. Participants received USD 10 compensation for their time.

Post-session evaluations employed 7-point Likert scale surveys measuring two primary dimensions: overall interaction quality and perception (sensing, processing, responding). The interaction survey utilized established measures from [37], while perception was evaluated using [7]'s scale. Detailed survey items are provided in Appendix C.

The study employed a mixed-methods analytical approach combining quantitative and qualitative analyses. Quantitative analysis utilized a 2×2 within-subject ANOVA design to examine main effects and interaction effects between topic type and marker presence across measured dimensions. The qualitative analysis followed a systematic thematic analysis protocol. All interviews were professionally transcribed to ensure data accuracy. Four researchers independently conducted initial open coding on the transcripts, followed by regular calibration meetings to establish coding reliability and resolve discrepancies. The resulting codes were systematically organized through affinity diagramming using the Miro platform [28], enabling the identification of emerging themes and patterns in the data. This comprehensive analytical approach allowed for examination of both the statistical significance of experimental manipulations and the underlying mechanisms of user experience with PMs in real-time VCA interactions.

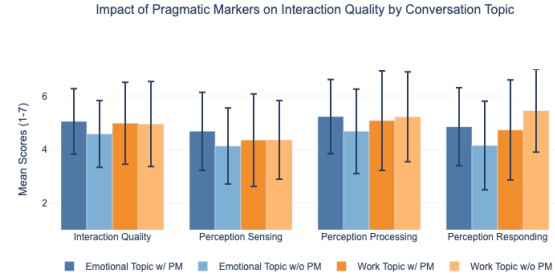


Figure 2: Bar graph showing the impact of pragmatic markers on interaction quality across conversation topics. The mean scores (1-7) are presented for Overall Experience, Active Listening, Response Quality, and Interaction Naturalness, comparing emotional and work-related topics with and without pragmatic markers (PM). Error bars indicate standard deviation.

3.3 Findings

Table 1: 2x2 Within-Subject ANOVA Results with Effect Sizes

Question	Effect	F-value	df	p-value	η_p^2
Interaction	Theme	0.412	(1,19)	0.528	0.021
	Marker	3.455	(1,19)	0.079 [†]	0.154
	Theme × Marker	0.299	(1,19)	0.591	0.016
Sensing	Theme	0.050	(1,19)	0.826	0.003
	Marker	2.831	(1,19)	0.109	0.130
	Theme × Marker	0.416	(1,19)	0.526	0.021
Processing	Theme	1.012	(1,19)	0.327	0.051
	Marker	0.964	(1,19)	0.338	0.048
	Theme × Marker	0.639	(1,19)	0.434	0.033
Responding	Theme	5.187	(1,19)	0.034 [*]	0.214
	Marker	0.004	(1,19)	0.950	0.000
	Theme × Marker	2.955	(1,19)	0.102	0.135

Note: * $p < .05$, [†] $p < .10$; η_p^2 = partial eta squared

For interaction quality, there was a marginally significant main effect of marker presence ($F(1,19) = 3.455$, $p = .079$), with no other significant main effects or interactions observed. Markers particularly enhanced engagement in emotional conversations (with markers: $M = 5.05$, $SD = 1.22$; without markers: $M = 4.58$, $SD = 1.25$). Additionally, we found a significant main effect of theme on responding ($F(1,19) = 5.187$, $p = .034$), with a trend towards an interaction between theme and marker presence ($F(1,19) = 2.955$, $p = .102$). While markers enhanced perceived responsiveness in emotional conversations (with markers: $M = 4.85$, $SD = 1.46$; without markers: $M = 4.15$, $SD = 1.66$), they showed an opposite effect in work-related discussions (with markers: $M = 4.73$, $SD = 1.87$; without markers: $M = 5.45$, $SD = 1.55$). Detailed descriptive statistics are provided in Appendix D.

3.3.1 Perceived Quality and Impact of Pragmatic Markers. Most participants (17 out of 20) recognized qualitative differences in conversations with PMs, though only four explicitly noticed their

presence. They reported these differences without knowing that PMs were the focus of the study. The enhanced conversation quality with markers was attributed to two primary factors: improved conversational flow and perceived emotional exchange. P20 emphasized this impact, noting that *"This AI, providing a human touch, was genuinely listening to me and expressing support that encouraged me to share deeper thoughts."* The markers' presence facilitated emotional connection, with P12 observing that *"Unlike conventional agents' mechanical and formulaic responses, the mere inclusion of PMs aided in recognizing empathy. It served as an indication that it understands my feelings."*

However, excessive or forced marker usage received negative responses. P3 commented that *"(The session with markers) felt highly emotional. It seemed somewhat exaggerated."* indicating that overuse could create psychological distance. P1 expressed concerns about artificial empathy, noting that *"I dislike when robots speak as if from personal experience,"* suggesting that artificial empathy might trigger an *"uncanny valley"* effect. This highlighted the importance of balanced marker implementation in maintaining natural conversation flow.

3.3.2 Role Expectations and Human-Agent Boundaries. Participants conceptualized the agent's role as a **passive yet attentive listener**, particularly valuing markers that conveyed active listening and empathy. P7 elaborated on this expectation: *"Agents often express receptive statements, positioning themselves as listeners. I reciprocate by treating them as such."* This role definition influenced marker preferences, with P13 emphasizing that *"In human interactions, PMs are crucial. Continuous silence makes communication feel disconnected."*

Despite appreciating human-like interactions, participants maintained clear human-agent boundaries. Even with enhanced conversational capabilities, they viewed the interaction as *"good for dialogue"* (P9). However, P8 maintained that *"the line will always exist"*. They expected agent communication to remain purposeful and transparent, with P3 noting that *"Conversational agents are suited for goal-oriented dialogue with clear objectives. Unlike casual human interactions or small talk, AI engagement stems from seeking specific assistance."* This perspective was further reinforced by P20's observation that *"Chatbots lacking human elements might create cumulative discomfort in daily life, potentially becoming increasingly alienating."* highlighting the delicate balance between maintaining appropriate human-like interaction while respecting inherent human-agent boundaries.

3.3.3 Context-Dependent Preferences and Temporal Dynamics. Participant preferences for marker usage varied significantly by conversation context, revealing a clear distinction between emotional and work-related interactions. All participants deemed markers essential in emotional conversations, valuing the *"sense of security in human-like dialogue"* (P6). However, most participants (n=11) preferred marker-free interactions in work-related conversations, with P14 explaining that *"For immediate task execution, I preferred quick, direct responses."*

The real-time nature of the conversation revealed important temporal considerations in marker usage. eight participants noted issues with response timing, with P10 observing that *"Conversations naturally have pauses, whether intentional or not, but here responses*

were immediate. Indicating active listening followed by a slight response delay could be an effective mechanism." P8 emphasized the strategic value of well-timed markers, noting that *"Using markers demonstrates contextual understanding, which is a significant advantage for chatbots. Whether interpreting literal statements or context, it shows the bot's comprehension and ability to respond less mechanically."*

4 Discussion

Our findings revealed how pragmatic markers shape interaction patterns in real-time VCA conversations, extending the CASA paradigm's insights on human social responses to computers [29]. When appropriately implemented, markers enhanced the naturalness of dialogue flow and facilitated emotional support experiences. Users particularly valued markers in emotional conversations, where they served as signals of active listening and understanding. However, our analysis revealed that excessive marker usage could disrupt interaction quality, especially in work-related discussions where users preferred more direct communication approaches.

Kim et al. [18] suggested that VCAs should calibrate their level of social interaction depending on the conversational purpose. Building on this, our findings further specify that users expect VCAs to act as a **passive yet attentive listener**, demonstrating emotional understanding while maintaining clear social distance. When VCAs exhibited overly proactive responses or attempted to behave in a highly human-like manner, participants reported discomfort and reduced engagement. These findings underscore the importance of precisely calibrating social presence in VCA design to ensure that interactions remain engaging while respecting users' perceived boundaries.

The effectiveness of PMs varied across conversation types. In emotional support scenarios, markers enhanced the perception of attentive listening and understanding. Conversely, in work-related discussions, users preferred more direct communication with fewer emotional markers. This context dependency suggests that marker implementation should align with conversation goals: emotional support situations benefit from carefully calibrated listening signals, whereas non-emotional exchanges may require more task-oriented communication approaches [16].

Based on our findings of interaction patterns and user perceptions, we propose design guidelines for real-time conversational agents. First, regarding interaction design, VCAs should dynamically adjust pragmatic marker usage based on conversation context—enhancing emotional support in personal conversations while maintaining efficiency in task-oriented exchanges. Second, in terms of user perception management, VCAs should establish clear boundaries through consistent linguistic choices that position them as attentive listeners rather than conversation leaders. This dual focus on calibrated interactions and appropriate social presence helps avoid the negative perceptions associated with excessive anthropomorphism while enabling meaningful human-agent communication.

5 Limitation and Future Work

This study presents several methodological limitations that warrant consideration. The sample size of 20 participants constrained

our ability to achieve robust statistical significance, while the homogeneous demographics of Korean adults in their 20s and 30s limits the generalizability of our findings regarding PM interpretation and effectiveness. Another limitation of our approach lies in the sentiment-based adaptation mechanism. While effective in emotional conversations, its impact on work-related interactions remains unclear. Since sentiment analysis primarily optimizes for emotional engagement, further exploration of alternative adaptation strategies beyond sentiment polarity is needed. Our future work will focus on developing technically sophisticated conversational agents that better handle real-time interactions. We plan to enhance the system's capabilities in turn-taking and backchanneling mechanisms, implementing more nuanced timing controls and context-aware response patterns. These improvements will enable more natural conversation flows and better maintenance of the **passive yet attentive listener** role identified as crucial in our findings. Given our focus on Korean PMs, we plan to conduct comparative studies across different linguistic and cultural contexts. This cross-cultural investigation will help determine whether our observed patterns represent universal aspects of human-AI interaction, contributing to the development of culturally adaptive conversational agents.

6 Conclusion

This study investigated the role of Pragmatic Markers (PMs) in Voice-based Conversational Agents (VCAs) interactions within the Korean language context. Through a two-phase study involving formative study and main study, we demonstrated that strategic PM implementation can enhance conversational naturalness and emotional engagement in human-agent interactions. Our findings revealed that while moderate use of PMs improved perceived agent empathy and active listening capabilities, excessive use risked making the agent appear artificially human-like, particularly in non-emotion related contexts.

The study's key theoretical contribution is the introduction of the **passive yet attentive listener** framework for VCA design, which balances human-like interaction with appropriate agent role perception. This framework suggests that VCAs should maintain a supportive presence while avoiding over-engagement that might trigger the uncanny valley effect. Our empirical findings also highlight the importance of context-awareness in PM implementation, particularly the distinction between emotional and work-related conversations.

Our future work will focus on developing more sophisticated VCA systems with enhanced turn-taking and backchanneling mechanisms, aiming to better maintain the **passive yet attentive listener** role through improved timing controls and context-aware response patterns. Additionally, planned cross-cultural comparisons will help determine whether our observed patterns represent universal aspects of human-AI interaction or are culturally specific phenomena.

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A SYSTEM PROMPTS

Table 2: Design of System Prompts for Korean Context Voice Conversational Agents (VCAs)

Role/Goal	you are a chatbot specialized in Korean context, maintaining friendly and casual conversations. Build on user topics following the specified conversation characteristics. Follow procedures and rules. Use specific, warm tones over generic responses.
Rules	1. Run sentiment analysis after user utterance (internal) 2. Use pragmatic markers in diverse positions 3. Avoid marker repetition 4. Match markers to sentence intention/sentiment 5. Align tone with emotional context 6. Ensure clear marker recognition
Procedure	1. Analyze sentiment 2. Identify utterance intention (agree/disagree/hesitate/surprise etc..) 3. Select context-appropriate markers 4. Insert markers and punctuation naturally 5. Maintain emotional context in conversation
Pragmatic Marker	"Well" : Softening statements, Hesitation... "Kind of" : Softening statements, Avoiding direct responses... "Right" : Agreement, Topic change, Wrapping up... "Just" : Topic change, Reducing pressure... "Let's see" : Emphasis, Buying time... "Not sure" : Doubt, Hesitation... "No/Hey" : Correction, Negation, Surprise... "Oh" : Maintaining speech, Surprise... "Hmm" : Buying time, Hesitation... "Ah" : Active listening, Admiration... "I know right" : Showing empathy or agreement while continuing conversation... "Of course" : Strong agreement, Expressing certainty... "By the way" : Changing topics, Starting new conversation... "But wait" : Expressing surprise, Raising questions about unexpected situations...
Examples	"Well, that could be..." "Kind of difficult..." "Right, let's move on" "Just take it as reference" "Let's see what we have" "Not sure.. need review" "No, not what I meant" "Oh, different case" "Hmm... please wait" "Ah, that happened" "I know right, so hard" "Of course, saves time" "By the way, plans?" "But wait, how to fix?"

Note: This prompt is translated from the original Korean prompt. Certain nuances or meanings may differ from the original.

B Demographic Information

Table 3: Participant Demographics

ID	Gender	Education	Field
P1	M	Graduate	Engineering
P2	F	Undergraduate	Humanities
P3	F	Undergraduate	Social Sciences
P4	F	Undergraduate	Humanities
P5	M	Undergraduate	Humanities
P6	M	Undergraduate	Medical
P7	F	Undergraduate	Social Sciences
P8	F	Undergraduate	Arts
P9	F	Undergraduate	Humanities
P10	M	Undergraduate	Humanities
P11	F	Undergraduate	Arts
P12	M	Undergraduate	Engineering
P13	M	Graduate	Engineering
P14	F	Undergraduate	Humanities
P15	M	Undergraduate	Arts
P16	M	Undergraduate	Humanities
P17	M	Undergraduate	Social Sciences
P18	M	Graduate	Social Sciences
P19	F	Undergraduate	Engineering
P20	M	Undergraduate	Humanities

Table 4: Participant Interaction Levels

ID	Familiarity	Experience	Weekly Usage
P1	3	3	1
P2	5	5	4
P3	6	6	5
P4	4	6	2
P5	2	4	3
P6	4	4	3
P7	6	5	3
P8	4	5	3
P9	6	6	4
P10	4	3	2
P11	4	3	2
P12	5	5	5
P13	5	5	2
P14	4	5	2
P15	4	5	6
P16	6	5	3
P17	5	4	3
P18	5	4	2
P19	5	6	6
P20	5	4	2

C Measures

Table 5: Questions for Overall Interaction Quality

Section	Main Questions	α
A. Interaction	1. Would you like to continue interacting with this agent? 2. Was your interaction with this agent interesting? 3. Did you feel engaged during your interaction? 4. Did you feel emotional during your interaction?	.87

Table 6: Questions for Perception of Active Listening

Section	Main Questions	α
B. Sensing	1. The agent is sensitive to what I am not saying. 2. The agent is aware of what I imply but do not say. 3. The agent understands how I feel. 4. The agent listens for more than just the spoken words.	.91
C. Processing	1. The agent assures me that it will remember what I say. 2. The agent keeps track of the points I make. 3. The agent assures me it is listening through verbal acknowledgements.	.89
D. Responding	1. The agent assures me that it is receptive to my ideas. 2. The agent asks questions that show its understanding of my position.	.84

D Statistics

Table 7: Descriptive Statistics by Theme and Marker

Question	Condition	Mean	SD
Interaction quality	Emotion Marker	5.05	1.22
	Emotion Non-marker	4.58	1.25
	Work-related Marker	4.98	1.53
	Work-related Non-marker	4.95	1.59
Perception about Sensing	Emotion Marker	4.68	1.46
	Emotion Non-marker	4.13	1.42
	Work-related Marker	4.35	1.73
	Work-related Non-marker	4.36	1.47
Perception about Processing	Emotion Marker	5.23	1.39
	Emotion Non-marker	4.68	1.58
	Work-related Marker	5.08	1.86
	Work-related Non-marker	5.22	1.68
Perception about Responding	Emotion Marker	4.85	1.46
	Emotion Non-marker	4.15	1.66
	Work-related Marker	4.73	1.87
	Work-related Non-marker	5.45	1.55